University of Idaho
College of Engineering

CS 4556/5556
PLC PROGRAMMING FOR AUTOMATION
FALL 2025

LECTURE 1: INTRODUCTION

INSTRUCTOR: DR JOHN SHOVIC
LECTURER: JACOB FRIEDBERG

WHAT THIS CLASS IS ABOUT

FALL 2025

- Programming Programmable Logic Controllers (PLC) for Automation
- Manufacturing, assembly, packaging use many different machines and robots
- They are all controlled by some kind of computer
  - Microcontroller – ESP32, ATmega (Arduino), ARM
  - PC - Embedded Windows, Raspberry Pi
- Enter the PLC
  - Going to program factories with ladder logic
  - Design Human Machine Interface (HMI) to communicate with Factory
  - Do some Python programming too

WHAT IS A PLC?

- An application specific computer
  - Processor
  - Memory
  - Storage
  - I/O
  - Networking
  - Expansion capability
- Robust
  - Solid state components
  - Can be hardened
    - liquid, dust, radiation, chemicals
  - Expected to run 24/7

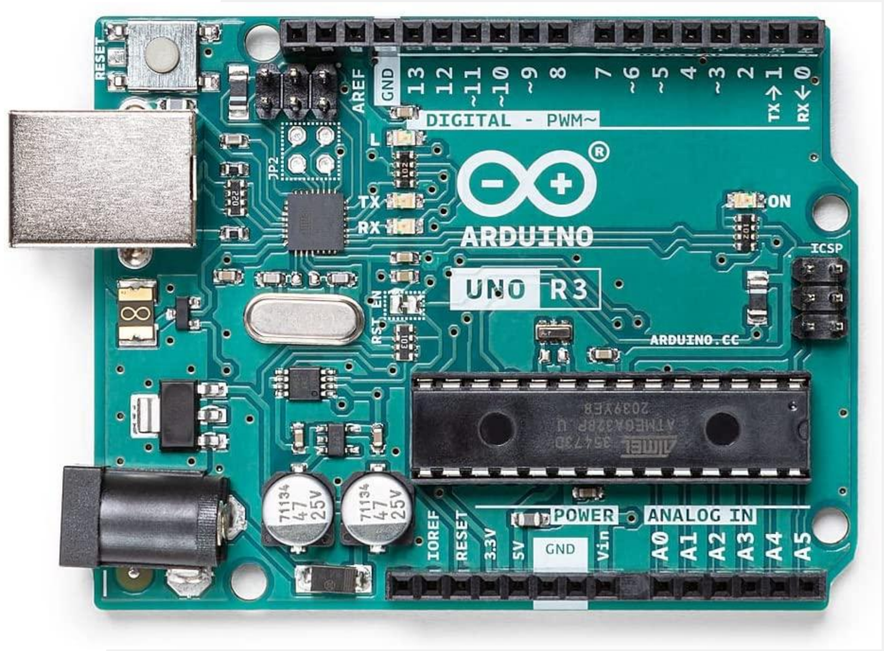






WHAT ABOUT OTHER DEVICES?

- Microcontrollers – small, fragile, not fast, passive cooling, low power, limited pins, limited storage, cheap
- General PC – larger, faster, active cooling, more power usage, easily expandable with PCIE cards, USB devices, storage, price varies





PLC, PC, MICROCONTROLLER COMPARISON

	Processor	Memory	Storage	I/O	Price
PLC	Varies	10-100K Words 3MB - 40 MB	10-100K Words SD Card	Serial, Ethernet, Pins,	$500-50,000
Desktop PC	3-5 GHz, 4-32 Cores	8-128 GB	GB-TB's SSD/HDD	Almost anything	$500-$5,000+
Raspberry Pi	1-1.5 GHz, 4 core	256 MB - 8 GB	GB's	Ethernet, WiFi, Video, Pins, Hats	$35-70
ESP32	80-240MHz, 2 core	320 KB	448 KB Flash	WiFi, Pins, Shields	$10-20
Arduino Uno	8-16 MHz, 1 core	2 KB	32KB Flash	Pins, Shields	$20-40

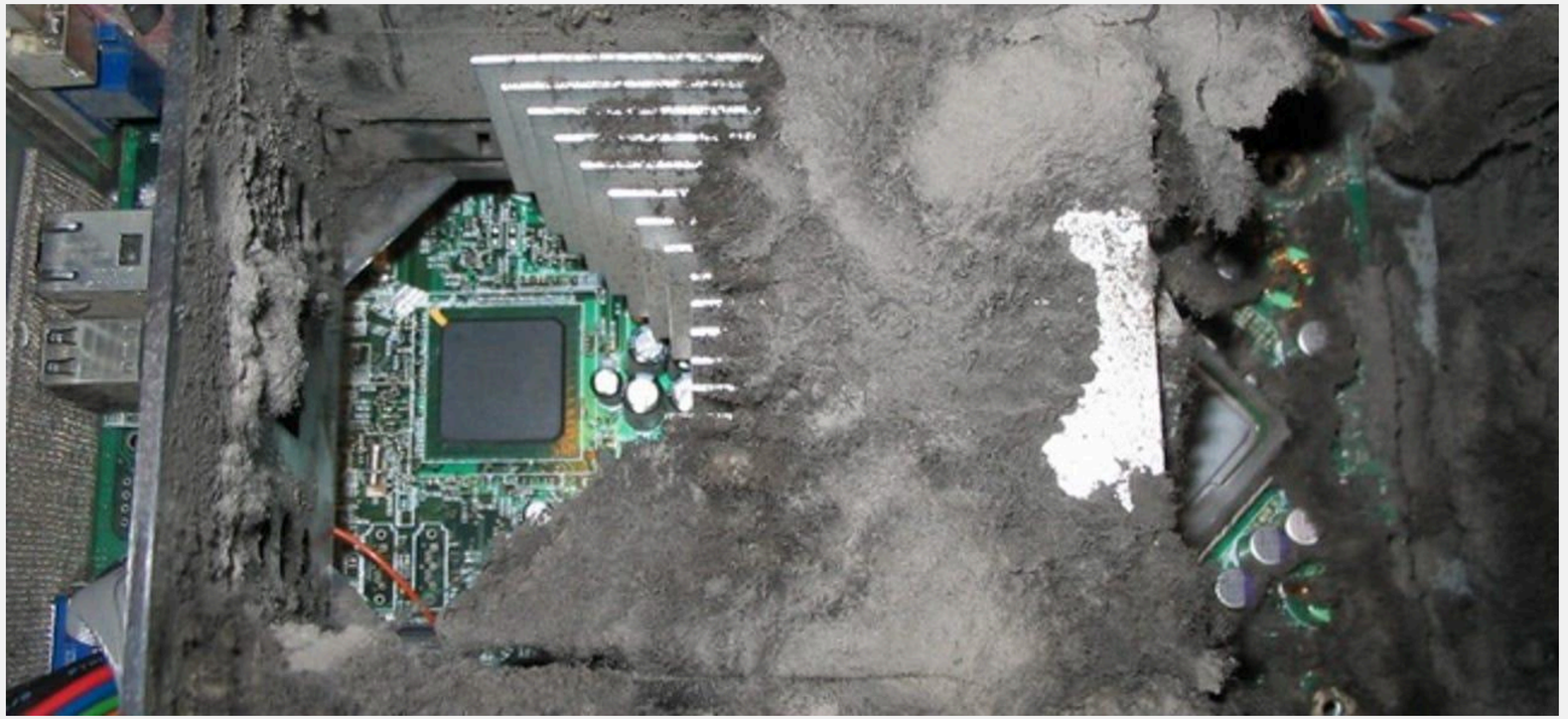
COMPUTER ARCHITECTURE – WORDS

- Words are a fixed(usually) length number of bits that a CPU architecture processes. Usually memory/addressing
- PLCs are defined in the IEC-61131-3 standard to have 16bit word sizes
- Modern computers x86-64 have 64-bit word sizes.
  - Microsoft defines(for backwards compatibility) 16bits as the word size
    - Half words = 8 bit also a byte
    - Word = 16 bit
    - Double words = 32 bit
    - Quad words = 64 bit
- Architecturally there needs to be hardware backed mechanisms to support word sizes and bit lengths.
  - Software can be used to emulate this, but at a significant performance cost due to having to spend extra cycles to only perform part of a computation.
    - 128 bit floats for example

- Word (computer architecture) - Wikipedia

FACTORY ENVIRONMENTS

- Noise (sound)
- Noise (electrical interference)
- Vibrations
- Impacts
- Dust, metal shards, wood
- Liquids
- Temperature
- Acidic/Basic Environments



PLC CABINETS

- Organized and labeled for maintained (Ideally)
- Cabinet Protections
  - Vibrations – no moving parts – Fans, HDD, DVD
  - Dust, liquid – no air intakes, fans, possibly high IP rating
  - Impacts – solid casing. Components are DIN rail mounted.
  - Wires secured in harnesses
  - Electrically grounded for safety of personal.

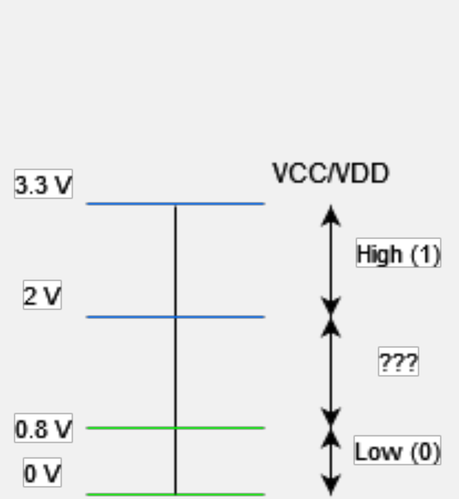


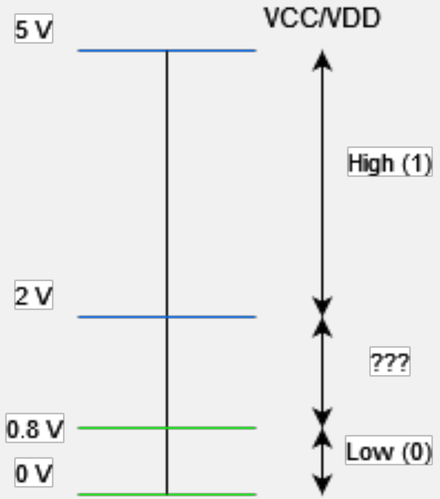
DIGITAL LOGIC – 0'S AND 1'S

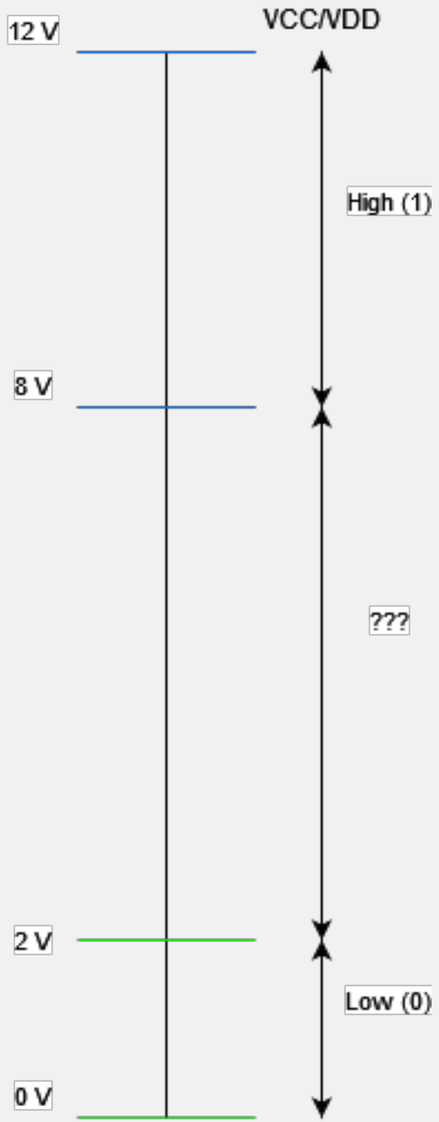
- 0 is low, 1 is high
- Voltage level is related to this
- Arduino Uno has a 3.3 V and 5 V model
- Map 0 V to 0 and 3.3 V / 5 V to 1
- PC has ~1 V for the CPU, RAM, 12V for fans, HDD, graphics card, 3.3 V for SSD motherboard, 5 V for USB
- PLC uses 24 V typically and sometimes 12 V, 48 V

WHY USE HIGHER VOLTAGE ON PLC?

- Voltage is continuous, not discrete
- Can take any value between 0 V and the high voltage
- Threshold values exist for both low(0) and high(1)
- In between, could be anything

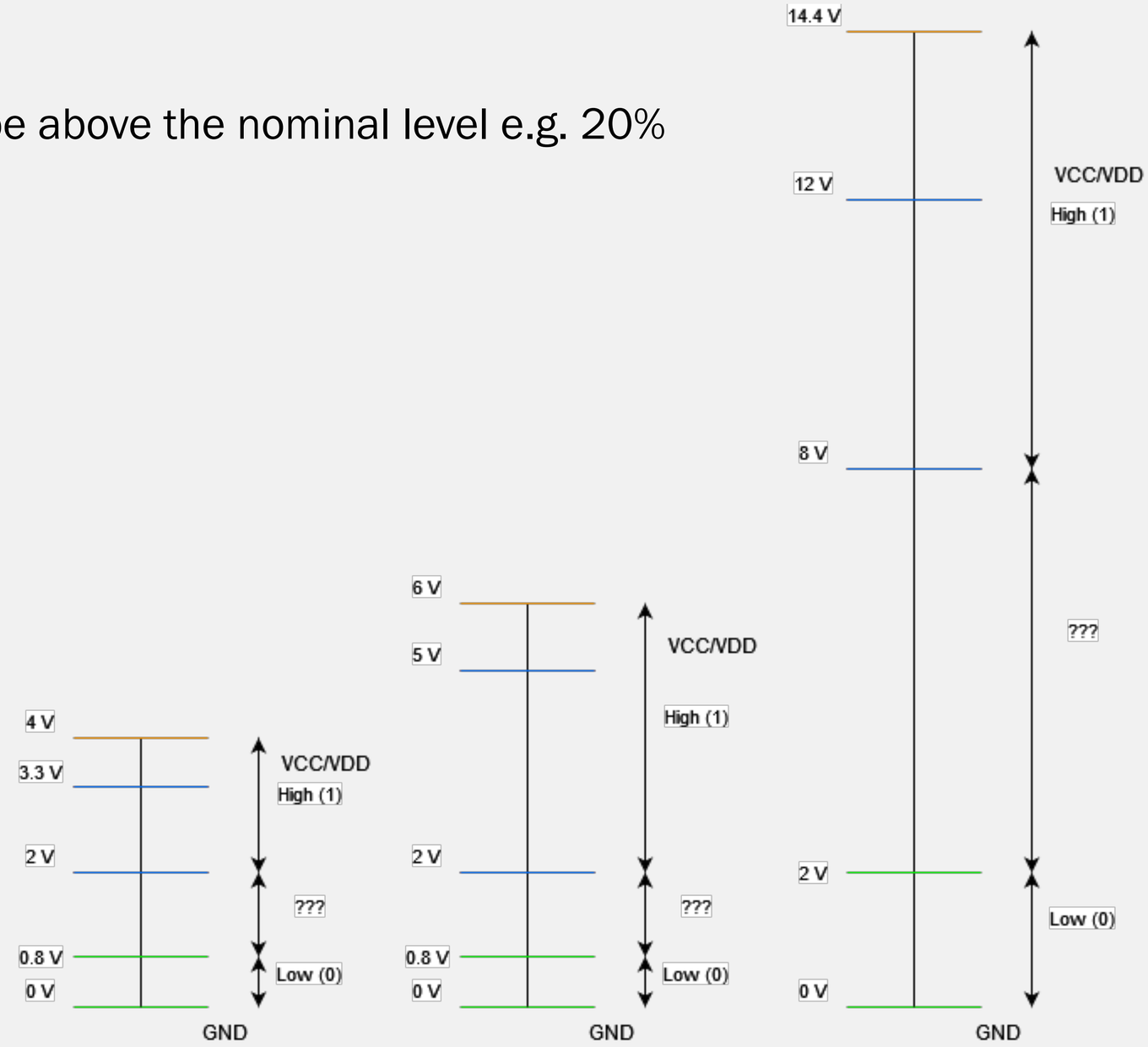






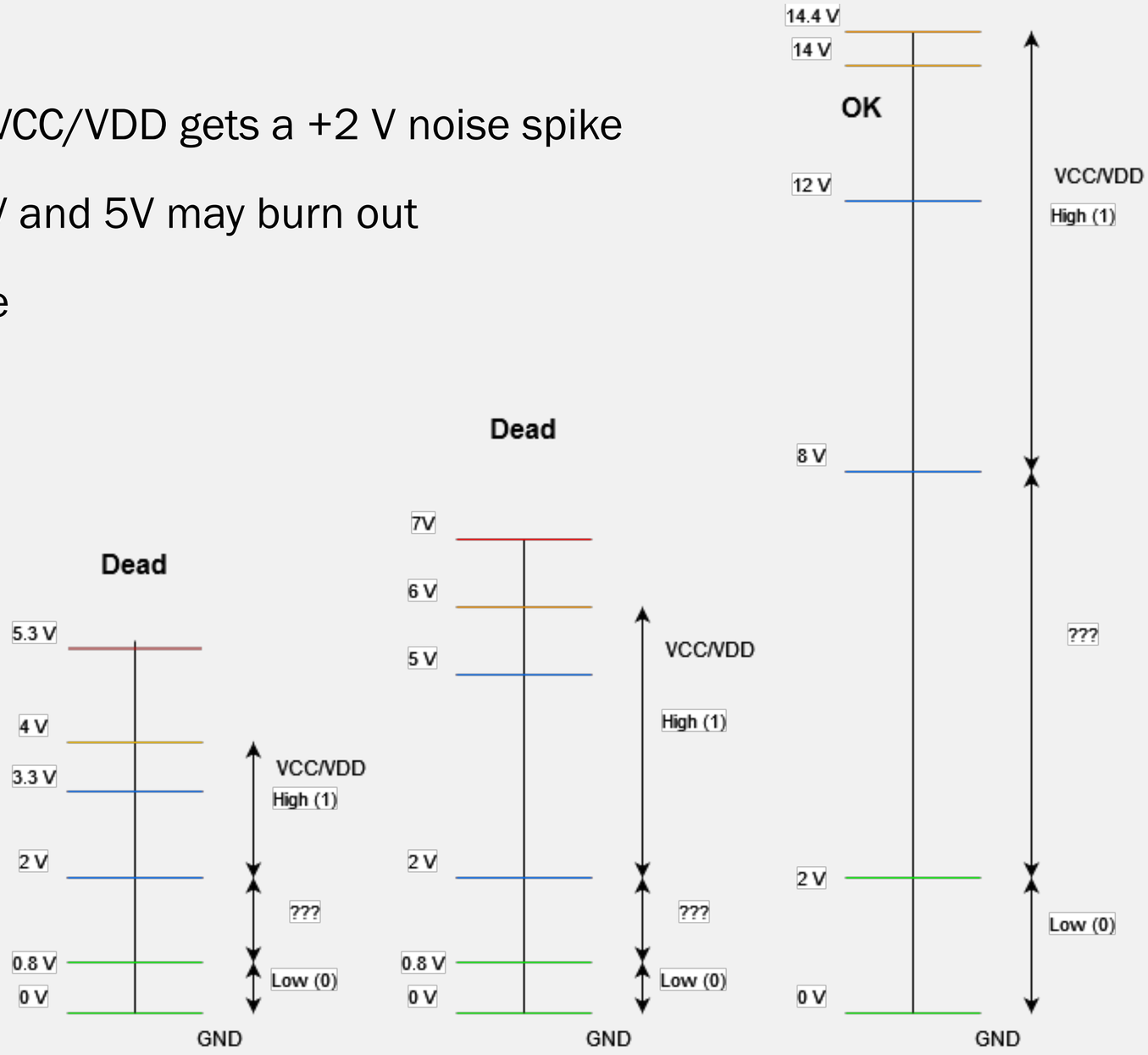
WHY USE HIGHER VOLTAGE ON PLC?

- Voltage level can be above the nominal level e.g. 20%



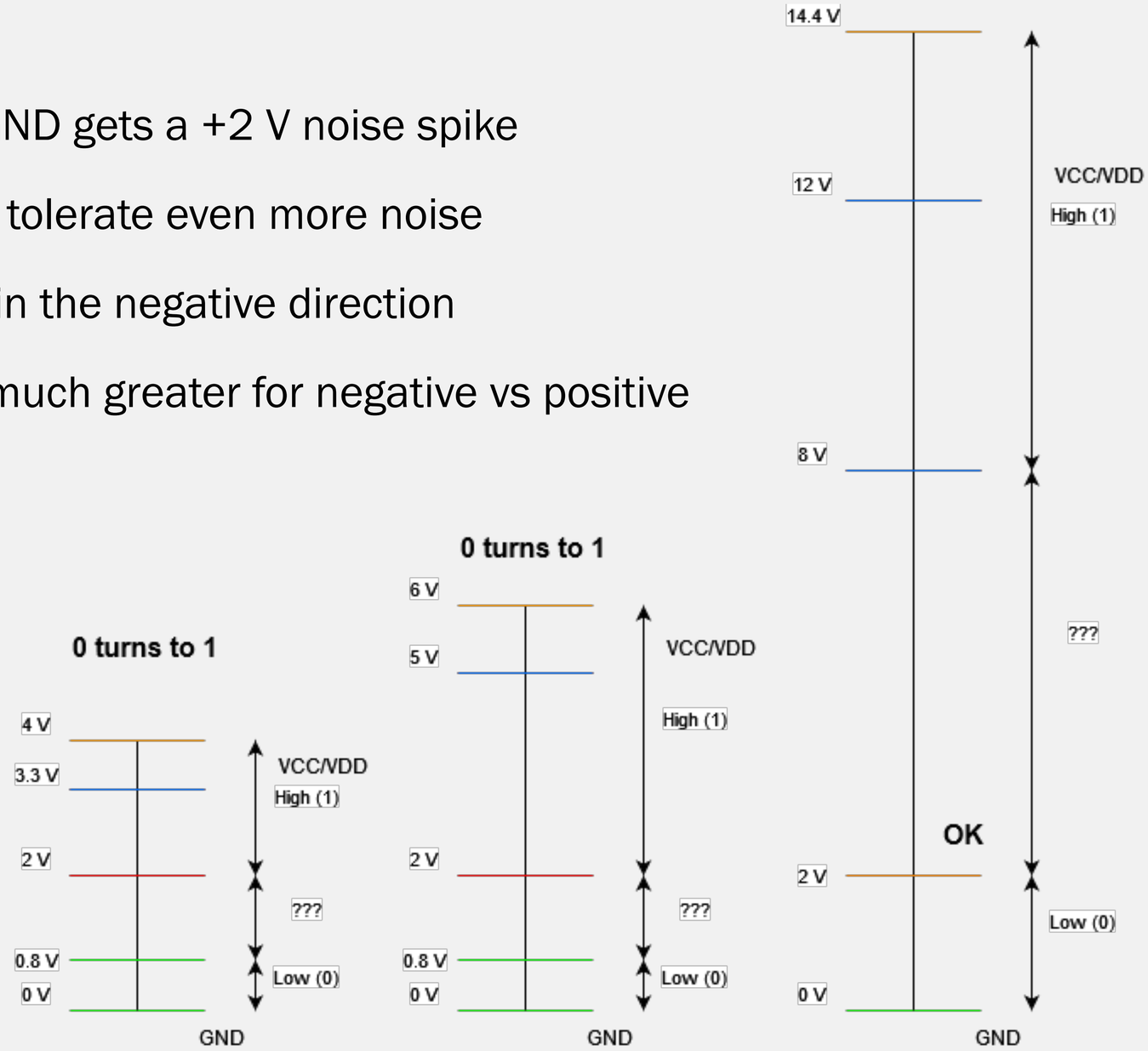
WHY USE HIGHER VOLTAGE ON PLC?

- The high signal at VCC/VDD gets a +2 V noise spike
- Transistors on 3.3V and 5V may burn out
- 12V has no change



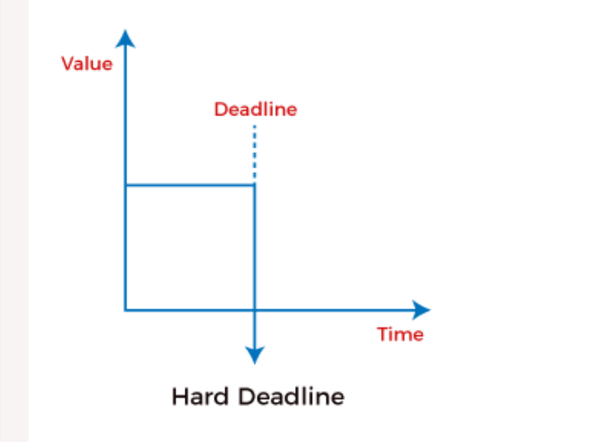
WHY USE HIGHER VOLTAGE ON PLC?

- The low signal at GND gets a +2 V noise spike
- 24 V and 48 V can tolerate even more noise
- Noise can also be in the negative direction
- Tolerance usually much greater for negative vs positive

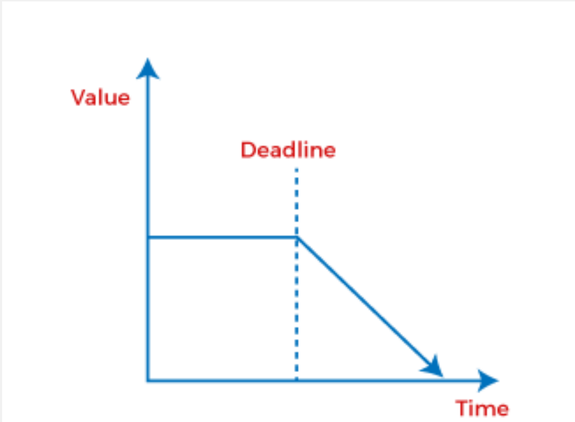


RECAP

- PLC's
  - Typically, 24v
  - 16 Bit word size
  - Purpose built rather than general computing devices
  - High environmental tolerances
  - Great for Hard Real-Time requirements.
- Digital logic is 1's(high, on) and 0's(low, off)
  - Threshold ranges for 1 and 0
  - Undefined region in the middle
- Two important take aways from why PLCs use 24V
  - Safe to work with (under ideal conditions)
  - Resilient to noise.



Hard Deadline



Soft Deadline